

UNIT 4 EMERGING NETWORKING TECHNOLOGY

- 4.0 Mobile Ad Hoc Networks (MANETs)
- 4.1 MANET Characteristics
- 4.2 MANET Limitations:
- 4.3 Advantages of MANET
- 4.4 Applications of MANET
- 4.5 MANET Architecture:
- 4.6 MANET-Operating Principle
- 4.7 Challenges facing ad hoc networks
- 4.8 Wireless Sensor Networks (WSN)
- 4.9 Applications of Wireless Sensor Networks (WSN)
- 4.10 Structure of WSN
- 4.11 Classification of WSNs
- 4.12 WSN Topologies
- 4.13 IoT introduction
- 4.14 Characteristics and Applications of IoT
 - 4.14.1 IoT in Home Automation: Intrusion Detection Use Case
 - 4.14.2 IoT in Smart Cities: Smart Lightening Use Case
 - 4.14.3 IoT in Environment: Air Pollution Monitoring Use Case
 - 4.14.4 IoT in Health: Wearable Devices Use Case
- 4.15 IoT System Architecture
- 4.16 Summary

4.0 MOBILE *AD HOC* NETWORKS (MANETS)

Today, we live in a digital age and are constantly surrounded by digital devices, such as laptops, mobiles, cameras, music players, etc. These devices are connected through various communication networks like WiFi, Bluetooth, Cellular Networks etc. In the earlier times, communication infrastructure had a fixed wired backbone with stationary cell towers and access points, as shown in figure 1. Such infrastructure networks, for example, cellular networks, were suitable for locations where access points were easy to install.

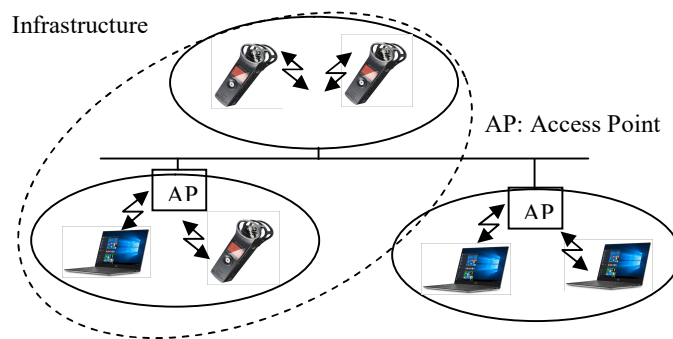


Figure 1: Typical communication between Infra. and APs.

However, today, the nodes (laptops, mobiles, etc.) are not always fixed at one point but can move in the system shown in figures 2, they connect to the fixed communication infrastructure of cell towers and communication access points (routers) further connected to the Internet backbone. However, networks can be configured without APs too. A collection of at least two or more electronic devices equipped with wireless communication and networking capability forming infrastructure-less networks without base stations is called a mobile ad-hoc network (MANET). MANET is a self-organizing and self-configuring multi-hop wireless network with a dynamically changing topology. The devices (nodes) in the network not only act as hosts, receiving data but also as routers that send data to other nodes in the network. MANET supports peer-to-peer communications and peer-to-remote communications with significantly reduced administrative costs. Therefore, the MANET can be defined as an infrastructure-less, fully distributed network in which each node acts as a transmitter, receiver, and router. The way communication is accomplished between the components present in the network is the main difference between the wireless and wired networks.

The term "Ad Hoc" network means a network that is temporary or implemented immediately when needed for a particular purpose. With the technological advancements in recent times, it is possible to use mobile nodes and mobile routers (a node can act as a router) to create a communication network where everything is mobile and nothing is fixed. In such a network, the topology of the network changes continuously as the neighbours of any given node are likely to change frequently. A few nodes will join the network at any point in time, while a few will disengage from the network, making the environment ad-hoc. With minimal or no dependence on infrastructure, a MANET is easy to deploy to support communication and computing anywhere, anytime.



Figure 2: Ad-hoc and Homogenous network

In heterogeneous networks, the devices have varying capabilities, while all the nodes have identical capabilities and responsibilities in homogenous networks.

4.1 MANET CHARACTERISTICS

- A MANET does not have a centralised infrastructure, unlike the traditional mobile wireless networks that need base stations, access points and servers to make it operational, as illustrated in Figure 1. Figure 2 shows a decentralized ad hoc network with all interconnected mobile nodes receiving data while functioning as routers simultaneously. One crucial characteristic of MANET is that it is a self-configuring network in which the network activities such as discovering the dynamic topology, delivery of messages and receiving data packets are executed by the nodes themselves.
- MANET has a dynamic topology in which nodes can freely move in and out of the network arbitrarily. It causes the network topology to change rapidly and unpredictably over time. For nodes to communicate in such dynamic

topologies, alternative paths are automatically discovered and data packets are forwarded across these multi-hop paths. Various route discovering mechanisms are used in MANETs to accomplish this.

- Another characteristic of a MANET is that it operates on lower bitrates with varying capacity wireless links. Compared to wired networks, a MANET has relatively low bandwidth, high bit error rate, and unstable and asymmetric links. Due to these factors, MANETs encounter higher delays and traffic congestion very frequently in comparison to fixed networks.
- A MANET is a network of nodes that are often hand-held devices with small battery backups. The battery-powered devices run out of battery support quite frequently. Thus MANETs are characterised by energy-constrained operations in which power conservation is essential. As a power conservation method, the network uses nodes in a manner so that they radiate power as little as possible and transmit data only when essentially required. Nodes can be configured to go into sleep mode.
- Compared to wired networks, MANETs have lesser physical security. As a result, they are vulnerable to various attacks such as eavesdropping, interception, denial-of-service and routing attacks. Therefore, different cyber security techniques are deployed to make MANETs more secure and robust against attacks.

4.2 MANET LIMITATIONS

The present and future need for dynamic ad-hoc networking technology is tremendous. This highly adaptive networking technology, however, still faces various limitations.

Throughput Drops with More Hops

Nodes in the transmission range of each other can send the data packets directly. However, a sequence of multiple hops is required when the node needs to send data packets to a non-neighbouring node. The data packets in such a case are forwarded to the destination node with the help of intermediate nodes acting as routers. With the increase in the number of intermediate hops, the throughput decreases rapidly.

Throughput Drops with Increasing Mobility

Highly mobile nodes result in increased overheads due to frequent topology variations. Suppose, in a situation of route failure, the need to discover new routes emerges and it results in enhanced demand for data packet retransmission. For example, in the proactive routing protocol, the routing table is used in which each node keeps a list of all available destinations and the number of hops required to reach each destination. The routing table also needs to change as the network's topology keeps changing. Any changes in routing table are then relayed to all the other nodes in the network, which significantly increases overheads in data transfer. Larger overhead leads to a lower percentage of the data packets reaching their destinations.

Delay Time

The time taken by a data packet to arrive at its destination after leaving the source node is referred to as delay. Nodes are busy with the transmission and receiving the packets to increase the network's throughput. Because of it, the queue at each node is always not empty or small which can lead to a longer delay.

History of Ad-hoc Networks:

The first generation of ad hoc network, called Packet Radio Network (PRNET), can be traced back to the 1970s when the Defence Advanced Research Project Agency (DARPA) launched packet-switched radio communication to provide reliable communication. PRNet provided an efficient means of sharing broadcast radio channels among many radios. PRNET, with low throughput (2 kbps per subscriber approximately), was not entirely infrastructure-less as it required a static station for routing. The PRNET then evolved into the smaller, cheaper, energy-efficient and cyber-resilient Survivable Adaptive Radio Network (SURAN) in the early 1980s. The United States Department of Defence (DOD) developed Globe Mobile Information System (GloMo) and Near Term Digital Radio (NTDR), providing a self-organizing and self-healing network. The NTDR, which uses clustering and link-state routing, was a widely used ad hoc network. The Internet Engineering Task Force (IETF) adopted IEEE 802.11 as the standard routing protocols for MANET for using mobile devices like PDA's, palmtops, notebooks, etc.

4.3 ADVANTAGES OF MANET

Compared to traditional networks, MANETs are cost-effective solutions and easy to set up, especially in geographical locations where fixed infrastructure is not feasible, as the nodes are not tethered. The device-to-device communication is used to create, implement and manage the MANET without any need for fixed hardware infrastructure. The ability to connect to the internet without any wireless router is the main advantages of MANET, making it more affordable than a traditional network. Their ability to handle connection failures is also created as routing and transmission protocols are designed to manage these situations. MANETs are extensively chosen due to their other distinct advantages, such as the ability to auto-configure, adaptive, self-diagnosing capability and cost-effectiveness.

4.4 APPLICATIONS OF MANET

The applications of MANET are not limited to a single area. There are many applications of MANET in industry, defence, medical and environmental sciences. A few of the applications include:

Tactical networks	Military Communication automated Battlefields
Sensor Network	Remote weathers for sensors, earth activities
Emergency Services	Disaster recovery, earthquakes, crowd control and commando operations
Educational Applications	Setup virtual class & conference rooms
Entertainment	Multi-user games, robotics pets
Location-Aware Services	Automatic Call forwarding, advertise location-specific services, Location-dependent travel guide.

4.5 MANET ARCHITECTURE

Depending on the application and the area to be covered, the MANET architecture may use body area networks (BAN), personal area networks (PAN), local area networks (LAN), metropolitan area networks (MAN) and wide (WAN) area networks as enabling technologies.

BAN coverage extends to about 2 m and is usually deployed using wearable devices. When the communication requirement is of the order of 10-15m, PAN is used. MANETs using LAN easily cover an area ranging up to 500m and are widely deployed for communicating within residential colonies, markets, or a cluster of buildings. WAN can also be used for even larger areas, although large networks have limitations in addressing, routing, and security management.

The MANET architecture layers, shown in figure 3.

- Link Layer
- Network Layer
- Transport Layer
- Application Layer

It is assumed that the appropriate physical layer is available.

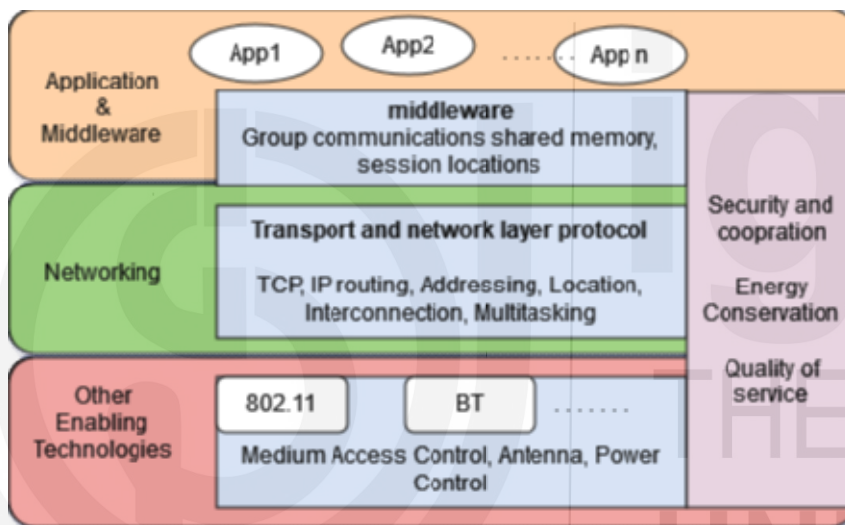


Figure 3: Reference MANET layer architecture

Link Layer:

The joint design of the link and physical layer can significantly enhance a MANET's efficiency and reliability. The link layer identifies priority packets and schedules packet delivery according to priority levels. It is accomplished through a Media access control (MAC) layer with a MAC discriminator and priority classifier. MAC is a protocol that enforces a methodology to allow multiple devices access to a shared media network.

The queue management section schedules the packets according to the priority levels. Data for real-time applications like live movies, video conferencing, etc., get higher priority than the packets for applications such as FTP and Email. When the network is congested, some packets may be dropped. Assigning higher priority to real-time packets ensures that they are not dropped in case of network congestion. Queue management thus helps in the timely delivery of data packets in real-time applications and improves the packet delivery ratio.

The MAC discriminator in the layer differentiates data packets that arrive from the wireless channel and sends them to the network layer. The address resolution protocol (ARP) packets go to the queue directly. The MAC packets used in IEEE 802.11 stay

in the MAC layer. The bandwidth estimation control packets are sent to the bandwidth estimation module for use in the routing layer's adaptive scheme.

The priority classifier classifies the data packets that arrive from the packet queue, whether required for real-time or non-real-time applications. Post classification, they are sent to the packet scheduler to schedule the packet delivery according to priority levels. This helps the distributed ad hoc network offer both real-time and non-real-time applications by granting real-time packets higher priority to capture the channel.

Network Layer: As described earlier, MANETs are self-configuring and have dynamic topologies with several addressing and routing issues. In large networks, the communication between nodes becomes unstable and the network itself becomes prone to cyber-attacks. For reliable, error-free data transfers, various protocols connect to enabling technologies to construct an end-to-end delivery mechanism between a sender and one or more receivers. The routing protocol should have provisions such as call admission or adaptive feedback to provide quality service. The network layer provides IP addresses to hosts and establishes routes between senders (sources) and receivers (destinations). For this, the network must map the dynamic topology and overcome the challenging addressing and routing issues. The network layer obtains the network resource information from lower layers and sends the network status to the applications layer. The main objectives of the network layer are to enhance the efficiency of the network by minimising overheads in routing, stability and reliability in computing routes with a high convergence rate, Congestion control, a better quality of service (QoS) and scalability by providing quality service in a large network with a large number of devices. The important parameters are delay and throughput.

Transport layer: Transmission control protocol (TCP) and user datagram protocol (UDP) are two transport layer protocols widely used in wired networks. In TCP, a connection must be established between communicating devices before the transmission of data. After transmitting the data, the connection must close. On the other hand, in UDP, there are no overheads for opening, maintaining and terminating a connection. In TCP, the delivery of data at the destination is guaranteed and it has extensive error checking mechanisms. UDP has a primary error control mechanism and the delivery of data to the destination is not guaranteed. UDP is faster, simpler and preferred for broadcast and has no congestion control mechanism to react to network congestion. Due to this, the applications using UDP as the transport protocol to transmit packets in MANETs can easily overwhelm the network with data. Such a scenario may result in considerable power wastage and limited use of available bandwidth in transmitting packets. TCP is comparatively slower and doesn't support broadcasting but has an inherent congestion control mechanism (not clogging the network and overloading the capacity in the routers). The standard TCP assumes that the end-to-end congestion is due to increased flow in the transport pipe and/or reduced available bandwidth. In wireless situation, congestion in TCP might be shown due to frequent breaks in the end-to-end TCP connection. Hence, the standard TCP needs to be modified and such modifications are available.

Applications Layer: The flexibility, ease of deployment and cost-effectiveness of MANETs make them a highly attractive option for several application scenarios. Based on the sensitivity to transmission delay, the applications are classified into real-time and non-real-time. Online live movies, live sports, and video conferencing are a few examples of real-time applications. On the other hand, applications such as Email and FTP are non-real-time applications.

Applications need to be designed to handle frequent disconnection and reconnection, time-varying delays and data packet losses. Initially deployed for the military operation, MANETs today are widely used in disaster management, educational, environmental, entertainment, law and order management and several location-aware applications. Extensively developed new communication technologies such as WiFi,

Bluetooth, IEEE 802.11, WiMAX and HyperLAN have accelerated the deployment of MANETs across several application domains. In addition, various applications developed for specific purposes form the application layer in MANET architecture.

4.6 MANET OPERATING PRINCIPLE

The nodes in an ad hoc network are initially unaware of the network's topology. They eventually have to discover the topology. During the discovery process, every node will learn about its neighbor nodes and the distance between them. This way, it also lets the other nodes know about its existence in the network. For efficient routing, routing tables are initialized and maintained by the routers using routing protocols and these tables are stored in their memory. The routers use the routing protocols to decide the path of the packet from the source node to the destination. Finding stable routes decrease the route-related overhead and finding the shortest routes are the main goals of the routing protocols.

- Fig. depicts a peer-to-peer multi-hop ad hoc network
- Mobile node A communicates directly with B (single hop) when a channel is available
- If a channel is not available, then multi-hop communication is necessary, e.g., A->D->B
- For multi-hop communication to work, the intermediate nodes should route the packet i.e., they should act as a router
- Example: For communication between A-C, B, or D & E, should serve as routers
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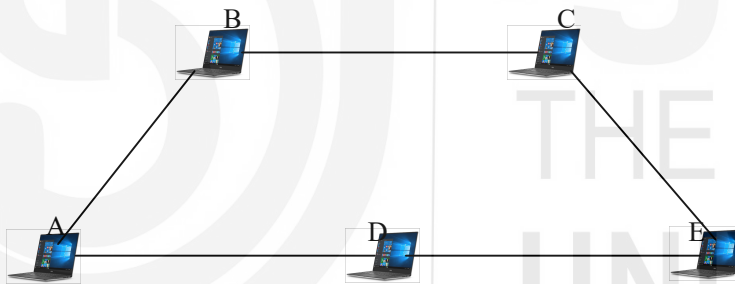


Figure 4: Example of an Ad Hoc Network

4.7 CHALLENGES FACING AD HOC NETWORKS

Regardless of the variety of mobile ad hoc network applications, there are still some issues and design challenges that we have to overcome.

MANET being a wireless network, it inherits the traditional problem of wireless networking:

- The channel is unprotected from the outside interfering signal.
- The wireless media is unreliable as compared to the wired media.
- The channel has time-varying and asymmetric propagation properties.

With these problems, there are some other challenges and complexities:

Bandwidth Constraints: The bandwidth of wireless links is much lower than its wired counterparts. For example, while several Gbps are possible in wired LAN, the wireless LANs typically work around 2 Mbps (nowadays up to 50 Mbps).

Energy constraints: The power of the batteries in the devices is the limiting factor that defines the operative time for the nodes.

High Latency: Nodes sleep or are idle when not in use to conserve energy. In data exchanges involving sleeping nodes, the delay might be higher if the routing algorithm needs to wake them up.

Transmission Errors: Attenuation and interferences are the effects of the wireless link that increase the error rate.

Scalability Concerns: Scalability is a crucial aspect of MANET, mainly when used in military communications. MANETs, when expanded, according to the need, each node must be capable of handling the expansion or intensification of the network.

Fault Detection and Management: MANET is an infrastructure-less network with decentralised administration. Faults become challenging to detect and manage as every node can communicate with every other node. In addition, dynamic topology, mobility and ever-changing routes add to the number of faults and data packet losses.

Security concerns: MANETs have more significant threats than the wired networks.

- Limited physical security: The physical sizes of the devices involved in these networks are so small that they can easily be victims of theft.
- Cyber security: They are vulnerable to cyber-attacks and can be compromised easily.
- Each node acts as a router and a receiver in the network, so each node plays two roles becoming significantly more vulnerable.

Check Your Progress -1

1.1 Military vehicles on a battlefield with no existing infrastructure will deploy network.

- (A) MANET (B) Cell Network
(C) LAN (D) Wi-Fi

1.2 What is the type of network in which the topology change from time to time?

- (A) Wi-Fi (B) Cell Network
(C) LAN (D) MANET

2. What is an ad hoc network?

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3. What are the challenging issues in ad hoc network maintenance ?

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4. Differentiate between cellular and ad hoc network in terms of cost effectiveness and bandwidth usage.

4.8 WIRELESS SENSOR NETWORKS (WSN)

As the name suggests 'wireless sensor network' is a network of sensor nodes. Now a question may come to your mind 'what are sensors'? Any hardware equipment which has the capability of sensing the physical characteristics of its surroundings may be called a sensor. To understand sensors, let's compare these with our body sensors, through which we sense our surroundings, smell by our nose, feel heat or cold by our skin, taste, etc. Similarly, if these senses are measured by any hardware device, we can say it is a sensor.

You must have seen the Fire Alarms installed in the buildings or in houses. Fire alarms have a sensor that continuously measures smoke or temperature around them and if the surrounding smoke or temperature increases by a certain limit, it starts beeping/ alarming. Generally, these fire alarms have wired connections, or they can have a wireless connection. So, can we put this fire alarm into the category of wireless sensor network? The answer is 'No' because generally, the fire alarms are a standalone system of sensors. It is not a network of sensors, and sensors are not talking to each other. However, we can create a wireless sensor network for the application of fire alarms in forests.

In the last decade, wireless sensor networks have gained popularity because of their applications and benefits. Some of the popular applications of wireless sensor networks are in Military, Agriculture, Security, and Healthcare. Let's discuss some examples of how wireless sensor networks have been changing the world around us.

4.9 APPLICATIONS OF WIRELESS SENSOR NETWORKS (WSN)

There are various applications of wireless sensor networks that we can see happening around us and many applications which are coming soon. In this section we will discuss some of these applications.

Agriculture: The deployment of wireless sensor networks in the field of agriculture has shown a lot of enthusiasm among the farmers. In agriculture, these networks are highly useful in monitoring and controlling the temperature, humidity, soil parameters, etc. So that farmers are getting better information and they can improve their production. In livestock management, wireless sensor networks have shown great relief to the farmers in monitoring the location and controlling the livestock and also their biological parameters.

Military: At present, we can see many wireless sensor network applications in the military. One of the most popular applications which many countries are currently using is protecting their border from any illegal trespassing or any intrusion attack. To do so, a network of sensors continuously observes the vibrations, sound, visuals, positions and many other physical characteristics in the border areas. If these sensors detect any suspicious activity, they can give this information to the nearby station or

to the military officers. Similarly, the military can do effective surveillance, intrusion detection, and collect intelligence with minimum risk on the battlefield in a better way by using wireless sensor networks.

Environment: There are a large variety of applications of wireless sensor networks in controlling and managing our environment.

- **Air or Water Pollution:** One application is monitoring the air pollution or monitoring the water pollution. We can have a network of sensors monitoring the pollution level in the air. If certain chemicals/air particles are increased more than a limit in the air it can provide us the information that pollution has increased in a particular location. Similarly, pollution management can be effectively handled in the rivers and oceans by sensing appropriate parameters.
- **Flood and Landslide Warning:** wireless sensor networks can be deployed on the bank side of rivers or in possible flood areas. These networks can give us a warning whenever the water level increases to a certain limit or water crosses into certain areas. Based on these warnings residents can be informed in a timely manner to evacuate the place. Similarly in the hilly areas, landslide warnings can be issued so that we can help residents to take the appropriate safety measures.
- **Forest fire:** As we discussed in the beginning regarding the fire alarm, we can create a wireless sensor network in a forest that continuously measures the temperature and the smoke level. Whenever there is a fire in some forest area the forest authorities can be informed about the fire, and they can take the appropriate measures to stop or the spread of fire.

Health: Wireless sensor networks are also useful in the health sector for effectively monitoring the patients' clinical data. We can provide various kinds of sensors to the patients to continuously measure their health parameters like a heartbeat, ECG, temperature, blood pressure, etc. These medical sensors keep recording the information from the patient's bodies and provide this information to a server. Server can analyse the information and assist doctors to make suitable decisions. These wireless sensor networks can be installed within the hospital where patients are admitted, or wearable sensors can be used so that patients can do their daily routine things and their clinical data can be recorded continuously.

Traffic: Another application of wireless sensor networks is in controlling and managing traffic. Various kinds of sensors can be installed in vehicles that continuously provide information to the base station so that the authorities can observe the number of vehicles in a particular area. Whenever there is a possibility of congestion in some location other vehicles can be informed or diverted to the other routes to avoid congestion. These sensor networks are also used to manage the parking lots and whenever there is any free parking area the vehicles can be directed to that location. Similarly, these sensors can also communicate from vehicle to vehicle so that any possibility of accidents is avoided.

Industries: Wireless sensor networks can be used within the industries to monitor the functioning of machines and if there is any change in the parameters such as temperature, pressure, or increase of chemicals then appropriate decisions can be taken for the effective working of machines.

There are applications of WSN in structural monitoring, like monitoring bridges and dams to reduce the human supervision and reduce the cost of monitoring, as well as improve the efficiency of the processes. As wireless sensor networks will grow; we will be able to see many more applications in the future.

4.10 STRUCTURE OF WSN

In this section, we will discuss a typical structure of wireless sensor networks. This structure will be helpful for you to understand the overall picture of wireless sensor networks and their components. You will be able to understand how a wireless sensor network is interconnected with the Internet or with the local network and how they are able to perform some desired actions.

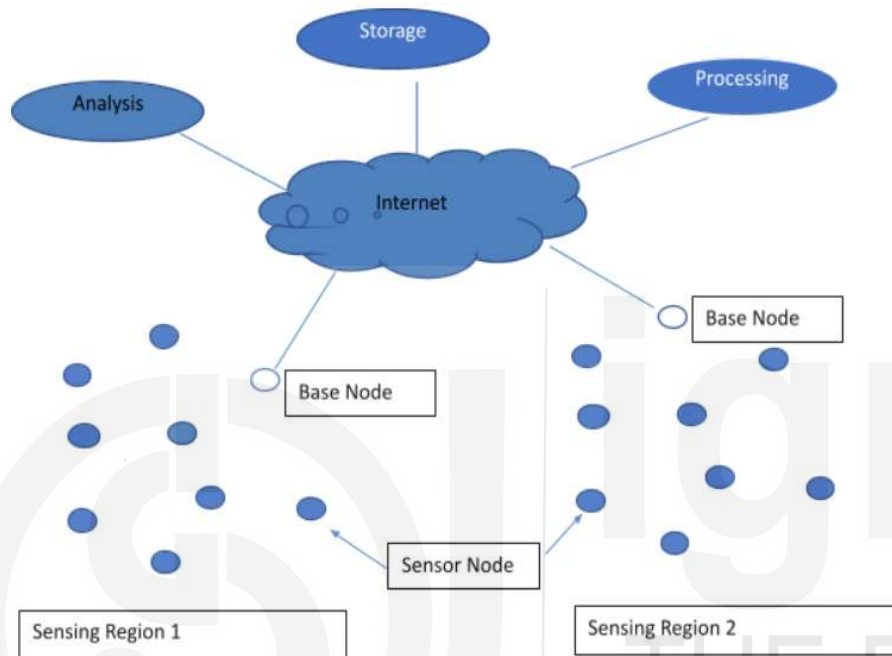


Figure 5: An example of WSN having two different sensing regions

The above figure 5 is showing an example of a wireless sensor network where we have two sensing regions. Each sensing region has a network of wireless sensors deployed in the region. All sensors in a sensing region are connected with each other through wireless connections. These sensor nodes communicate with each other and share the data with others and also communicate and share the data with Base Node or Base Station.

The purpose of base node is to give command to the sensor nodes to perform certain tasks of sensing, based on the command given by the base node all sensors collaborate with each other to send back the required data. Base node is also responsible for accumulating the data from the sensors and passing it to the Internet or a specific network. Further, this network or Internet is connected with different computing units, which store, process and analyse the data coming from the base nodes and perform the necessary action/decision.

Sensor Node

A sensor node is a basic component of wireless sensor networks. Each sensor node has a power unit, sensing unit, communication unit and processing unit. These sensors are radio frequency-based sensors which can transmit / receive data to/from a short

distance. Hence, each sensor node also works as a relay to pass-on the data to other nodes. Let's see the structure inside a sensor in the following diagram.

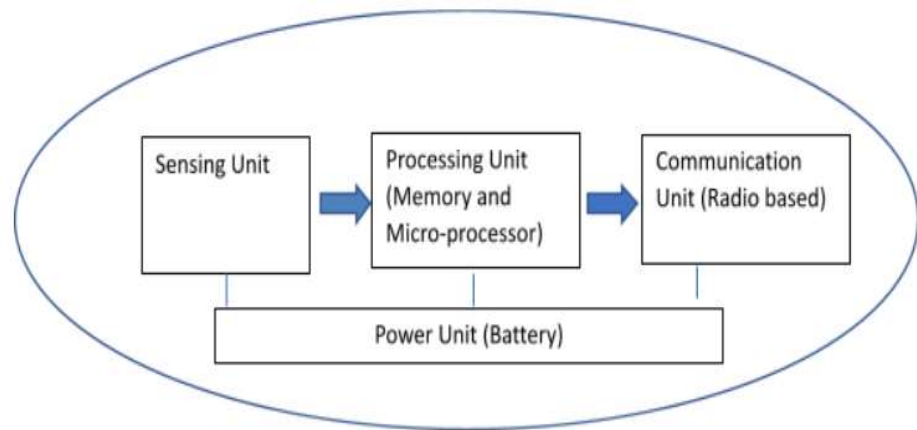


Figure 6: An example of a Sensor node structure

There are different types of sensors available at present which can easily sense physical characteristics around it like temperature, pressure, vibrations, sound, flow, humidity, radiations, motion, position, light, etc. Also, many other types of sensors are under development for various purposes which will be available for the desired implementations.

As depicted in the above diagram, a sensor has a sensing unit which transfers its data to the processing unit, this data is processed and stored by the processing unit and further it can be communicated to the base station and/or to the nearby sensors. The power unit is generally a battery-based power unit which provides the required power to all these above units of a sensor.

4.11 CLASSIFICATION OF WSNs

Wireless sensor networks are broadly classified in following types:

- PAN, LAN, MAN, WAN wireless sensor networks
- Stationary and Mobile wireless sensor networks (WSN)
- Single base and Multi-base Station
- Single Hop and Multi-Hop WSN
- Self-Organised and Non-Self-organised
- Homogeneous and Heterogeneous WSN

Let's understand the differences and similarities between these WSNs

PAN, LAN, MAN, WAN wireless sensor networks

This classification of wireless sensor networks is based on the size of wireless sensor network. You might have heard about these terminologies before. Let's refresh the

meaning of PAN, LAN, MAN and WAN. PAN refers to a personal area network, some typical examples are Bluetooth, and wireless USB. Any wireless sensor network which is working in a very short area within a personal area network is a PAN wireless sensor network. Similarly, depending on the size of our WSN we can classify these WSN into LAN (local area network), MAN (metropolitan area network) and WAN (wide area network) WSNs.

Stationary and Mobile WSN

Stationary sensor network is a network of stationary sensors placed in a particular region and they communicate with others and pass the information to the nearby base station from a fixed physical location. However, in Mobile wireless sensor networks the sensors are not stationary and keep changing their physical location e.g. sensors installed in the vehicles.

Single base and Multi-base Station

As we have seen earlier in our example of a wireless sensor network with two regions that had a single base station for each sensing region which is giving command to sensors and communicating with the network. However, in multi-base station scenario, we are having two or more base stations to give commands to sensors in a sensing region and to collect the information from these sensors.

Single Hop and Multi-Hop WSN

In a single hop wireless sensor network, all sensors are directly connected with the base station and sensors are directly receiving command from the base station and sending back the required information. However, when we have very large sensing regions that time this one-to-one connection between base station and sensors is not very effective and it may consume high energy. That's why, each sensing region is further divided into different clusters and all sensors are also divided into clusters. One sensor in a cluster will be treated as a cluster head or a supervisor sensor. Now, in this arrangement the base station will communicate with the cluster heads and sensors in a cluster will communicate with the cluster head. All cluster heads will send the required information to the base station, not the sensor nodes. This kind of arrangement of sensors with cluster heads to Base Station is called multi-hop WSN.

Self-Organised and Non-Self-organised

A non-self-organised wireless sensor network is a type of system where sensors are not capable of organising themselves in a network and needs a control unit to accomplish a task given by a base station. However, nowadays most of the wireless sensor networks are self-organised networks, they can organise themselves in a network and collaborate with each other to accomplish the task given by the base station.

Homogeneous and Heterogeneous WSN

In our previous example where we discussed the components within a sensor. We have seen that each sensor has sensing, processing, storage and communication capabilities. If all sensors are having the same type of sensing, processing, storage and communication capabilities we can say that as Homogeneous WSN. However, if all sensors are not having the same sensing, processing, storage and communication capabilities the WSN is heterogeneous. Some sensors have high capabilities, and some are low. In that case, we will call that Heterogeneous WSN.

4.12 WSN TOPOLOGIES

In this section we will discuss different topologies of wireless sensor networks, which means how these radio frequency sensors and base station(s) can be arranged as a wireless network.

Star Topology WSN

This is one of the most common kinds of wireless sensor network, where all the sensors are connected to a base station and this base station is commanding all the sensors and receiving the data provided by the sensors.

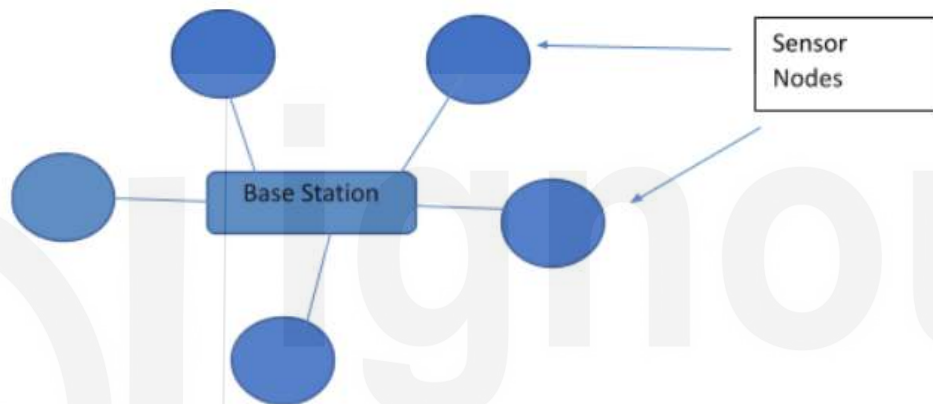


Figure 7: Star Topology of WSN

one advantage of star topology is its simplicity to implement, second this arrangement won't add much power consumption on the sensors, third the communication delay between sensors and base station is minimal. However, in terms of demerits- the base station should be in the radio frequency range of all the sensors so that sensors can send and receive data.

Mesh Topology WSN

In mesh topology all sensor nodes can communicate with each other, this kind of topology is useful in multi-hop WSN. If one node wants to transmit a data to another node which is out of its radio frequency range then one node can transfer the data to the nearby node and that node can transfer data to the destination node, this way, we can achieve a multi hop wireless sensor network. The main advantage of the system it's that it can be deployed in a very large geographical location, furthermore, mesh topology is more reliable because if some node is failed to transfer any information, then other nodes can be used as a relay within a network. The main disadvantage of this arrangement is the increased power consumption by the sensor nodes.

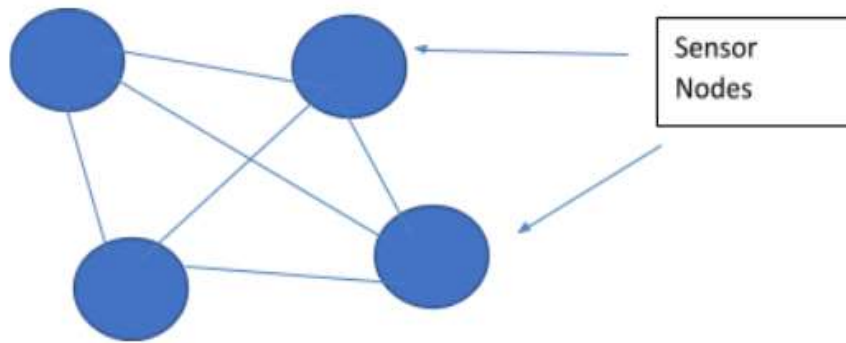


Figure 8: Mesh Topology

Also, Mesh topology can be integrated with star topology and form a hybrid topology, which can have advantage of both the topologies in Wireless sensor network.

Check Your Progress -2

1. What is a Wireless sensor network?

2. Write short notes on Sensor node.

3. Differentiate between single hop and multi hop WSN.

4. Name different topologies of WSN.

5. Give an example of a heterogeneous WSN.

4.13 IOT INTRODUCTION

IoT is a network of physical objects, called “Things”, embedded with hardware like - sensors or actuators or software, for exchanging data with other devices over the internet. With the help of this technology, it is possible to connect any kind of devices

like simple household objects for example- kitchen appliances, baby monitors, ACs, TVs, etc to other objects like- cars, traffic lights, web camera, etc. Connecting these objects to the internet through embedded devices, allows seamless communication between things, processes or people.

Some of the applications of IoT devices are – smart home voice assistant Alexa, smart traffic light system. IoT devices when connected to cloud platforms can provide a huge and wide variety of industrial or business applications. As the number of IoT devices are increasing, the problem of storing, accessing and processing is also emerging. IoT when used with Cloud technology provides solutions to these problems due to huge infrastructure provided by the cloud providers.

Internet of Things (IoT), is also seen as an extension of the Machine to machine (M2M) communication with some more capabilities like sensing, acting and to some extent decision making. Here the large network of small devices communicates via cloud or other networking platforms. It enables the users to communicate/ control a large number of devices in a minimum time and to the maximum extent. Hence IoT as a system comprises of **smart objects** having components like network, sensors, power, processing power, data and services around it.

4.14 CHARACTERISTICS AND APPLICATIONS OF IOT

Characteristics and Applications of IoT

The participating devices in IoT are required to have few characteristics like communication, power, sensing & actuating (not necessarily) and data processing (to some extent). Many of the devices also have some processing and decision making capabilities and it depends on the requirement of the system. These all are characterized in four different classes of fundamental characteristics namely heterogeneity, dynamic, interconnected and scaling.

- **Heterogeneity:** Different type of device participates in the network, here hardware platforms and communication networks also differ but at the end they all provide a seamless connectivity of the thing.
- **Dynamic:** The smart things under monitoring change their states, orientations very frequently. Sometimes the number of participating/ enabling devices is also dynamic (with self configuring capability) and hence the entire network changes on the go.
- **Interconnection:** The aim of the IoT is to make things interconnected to the outer/ digital world and make things accessible/ controllable from the remote location as well. With IoT, theoretically, you can connect anything and everything which is of some use for you.
- **Scaling:** As number of things around us is enormous in their connection and other related devices would be more in number. This number increases with the increase in the things of our interest. Hence the IoT devices/ network have to be scalable in nature.

As we are now aware about the IoT as an intelligent connected network of smart things/devices, we can see the following application of the IoT in our real life.

- Monitoring industrial processes, equipment health and product quality.
- Monitoring the health of smart buildings, critical infrastructure and vehicles.

- Environment monitoring and data collection in extreme climatic conditions.
- Patient monitoring at homes as well as in emergency vehicles, and OTs in hospitals.
- In many other industries like automotive, aviation, retail, agriculture, defense and space explorations.

Apart from the above mentioned scenarios, there are several other application of IoT where a smart device with some power, sensing capability and communication capability is required.

Now you have a basic understanding of the characteristics of the IoT devices and their purpose of use. Let us now discuss few of the use cases where IoT can help users in many ways. We have a number of use cases in various sectors of which a few are mentioned here as examples.

4.14.1 IOT IN HOME AUTOMATION: INTRUSION DETECTION USE CASE

Security of the home is a major a matter of concern to everyone especially when nobody is at home. It is also important to ensure the security of children and elderly members of the house. A lot of security solutions are available today from access control to intrusion detection when an unauthorized person tries to enter in the house. IoT has made these systems work smartly and effectively in a real time mode.

A typical home intrusion detection system makes use of smart cameras with night vision and motion detection capabilities. Some of the intrusion detection systems also use Passive Infrared (PIR) and door sensors to detect the motion and inform the users. Many of such systems also send short clips or images of the intruders to the users in very short/ real time. These systems can also be controlled using cloud based platforms and also possess on-system memory to store some vital information. The below given figure 9 represents the typical setup of the intrusion detection system for a house.

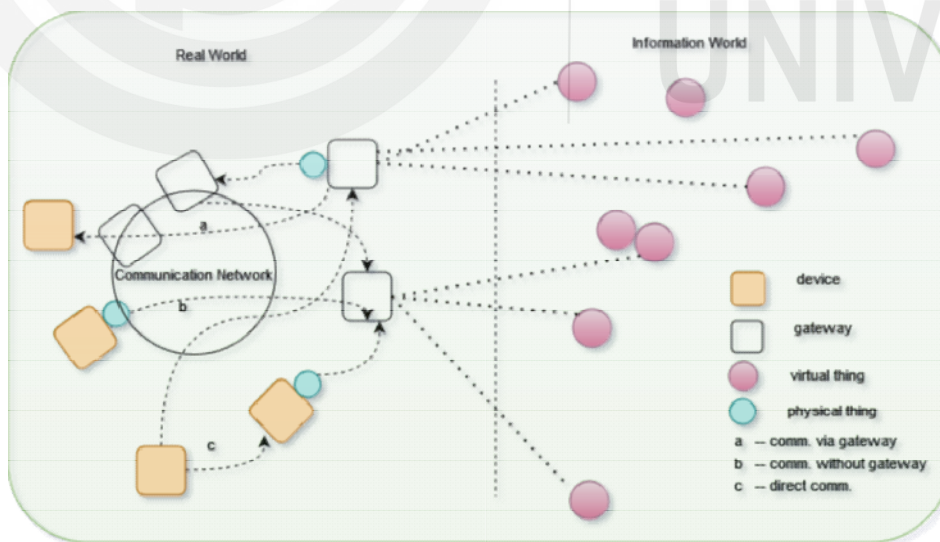


Figure 9: Home automation, intrusion detection systems

Here in the figure, the door sensors or motion detection sensors are connected to the gateway wirelessly. Sensors come with wireless connectivity like Bluetooth, WiFi or

ZigBee. The smart camera system often comes either with WiFi or Ethernet connectivity. The gateway shares the collected information to the cloud based analytical engine which processes the real time information and makes relevant decisions. The users get appropriate notifications/ alerts about the decision made by the analytical engine. These engines can be trained/ configured for the user specific requirements and hence provide security to the home.

4.14.2 IOT IN SMART CITIES: SMART LIGHTING USE CASE

Lighting in homes, buildings, roads and parks is one of the major areas of energy consumption. Majorly the electricity is produced using non renewable energy sources worldwide. These include petroleum products like oil and gas, coal and other resources. These resources are not in abundance and also contribute to environmental pollution in many ways. The more demand of electricity increases the consumption of these resources. Apart from some essential sectors, many sectors can contribute to energy saving strategy to save electricity. IoT enabled devices and smart monitoring of consumption helps in saving electricity. Today, we are focused on developing more energy efficient green buildings to save our environment and lower carbon footprint. The concept of green building itself incorporates the energy efficient and environment friendly ecosystem. IoT enabled devices helps in scheduling the lighting and controlling light source intensities depending upon the ambient conditions and requirements and hence reduce energy wastage in these buildings. These devices can also be controlled at the remote locations by the authorized operator/ persons. Sensors on these devices also gather environmental data and device parameters. This helps in better monitoring of the working of devices in these environments and improves the device life. These devices also monitor the building's structural health and monitor indoor environment health. Smart street lighting, park lighting and condition dependent lighting (like in fog or in storm) are some examples of IoT use case for smart lighting.

4.14.3 IOT IN ENVIRONMENT MONITORING: AIR POLLUTION USE CASE

Small size of devices and sensors with minimum battery/ power requirements are the key advantages for any air quality monitoring sensor/ device. These can be placed at desired places with ease and can monitor the air quality. A typical air pollution monitoring IoT devices consists of a few sensors for monitoring CO, CO₂, NO, SO₂ and other poisonous gas levels in environment. These are also equipped with GPS (Global Positioning System) system for location and wireless transmission of data to the cloud. In the cloud, analysis of the information received is carried out and outcomes are reported to the users/ authorities. Every year during the winter season many Indian states face severe air pollution condition. Various factors contribute towards it and regulatory policies are implemented. IoT devices have contributed it a lot for air pollution monitoring. Using IoT devices, government collect all the required data at one go from different places and governments can issue guideline for the citizens accordingly. The constituents of the pollution can also indicate the industry sector which is responsible for majority of the pollution at a particular place. The authorities can take better and fast decisions to tackle the situation. On the other hand, various air purifiers are being made to address the issue in a closed door environment. This contribution of IoT devices can make authorities swiftly issue

guidelines for residents if higher levels of air pollutants are reported in the nearby areas.

4.14.4 IOT IN HEALTH: WEARABLE DEVICES USE CASE

Health monitoring is major area where IoT is contributing. Many smart gadgets are now available in market which monitor vital body parameters and logs them in the personal devices. We are recording our pulse rate, oxygen saturation, ECG, temperature and many other body parameters which are majorly non invasive in nature using IoT devices. These devices are small, and battery powered. These devices collect body parameters and store the data either at the local storage or in the cloud where Artificial Intelligence based algorithms extract useful information which can be shared with the doctors for seeking any medical advice. Various health devices are now capable of making emergency communications to family members or health officials/ doctors in any emergency situation. These devices are also useful in tracking the school going children or elderly people who require extra safety and attention.

4.15 IOT SYSTEM ARCHITECTURE

IoT connects physical things to the network and hence various network devices are involved. The physical things are made smart by placing some sensors on them to sense their state, orientation or environment. These sensors require power modules to operate and communication modules to send the sensed data. Communication can be either in wired mode or in wireless mode. As an example we are here considering the wireless mode only. This includes WiFi, Bluetooth, ZigBee, LORA and many other technologies. A typical system architecture of IoT for physical world network is given below in Fig.

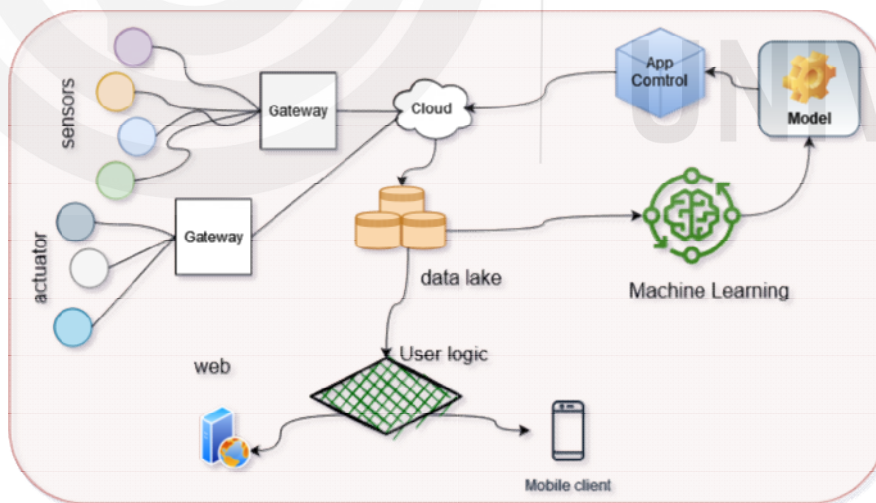


Figure10: Typical IoT communication

The given figure is self explanatory and you can make out the communication of data from the flow of information signals shown in the figure 10. The figure also shows the communication via gateway, without gateway and device to device direct communication.

There are many protocols that IoT devices use for communication and message sharing on the network. A few of the IoT protocols are shown in the figure 11 given below. These protocols work at different layers of the protocol stack like Application, Transport, Network and Link layer. The devices which connect real world things to the network are small and battery powered (most of the devices). These devices communicate with each other or other networking infrastructure for data exchange using various communication protocols. The below given figure shows the various protocols used by IoT devices at different layers.

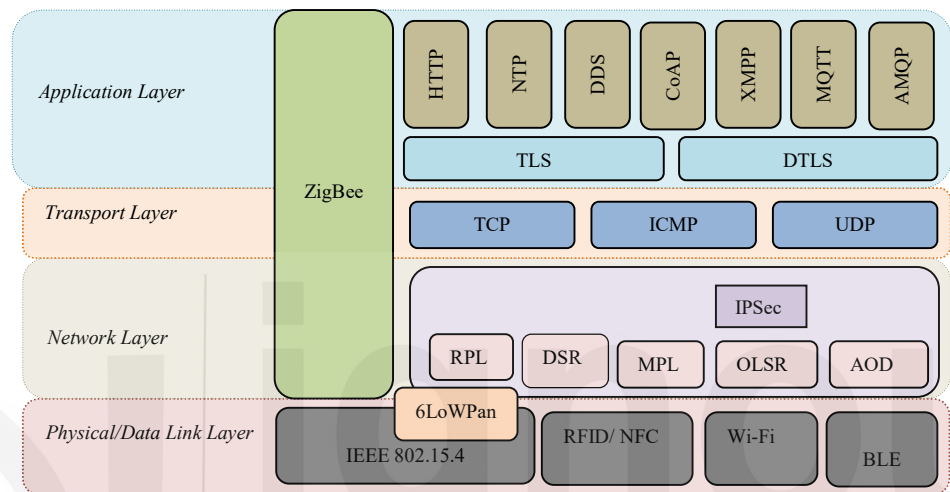


Figure11: IoT protocols

NOTE: The details of these protocols are given in a later course MCS-221. You are advised to refer MCS-221 for further details.

Check Your Progress -3

1. What are things in IoT? What are characteristics of IoT ?.

2. How IoTs communication differs from M2M communication?

3. Write short notes in ZigBee, NFC and Bluetooth.

4.16 Summary

In this unit we have learnt about different emerging technologies in networking namely MANET, WSN and IOT. Details about the architectures, usage, advantages and disadvantages are given. We have also learned about the various scenarios of using these technologies in day to day life. In MANET, characteristics and operating principles are described in detail. The structure of wireless sensor networks with its types and classification is also mentioned. WSN topologies like star and mesh are discussed in detail. AT the end, IoT and its network architecture, Things and smart objects are mentioned as integral part of the IoT network.

Solution to Problems

Answers to CYP-1

1.1 Answer: A

1.2 Answer: D

2. An ad hoc network is a multi-hop, infrastructure less network which has no centralized server to control the communication between the nodes and resources cannot be reserved beforehand. It is used in battlefields and military applications.

3. Following are the challenging issues in ad hoc network are

- a) Medium access scheme
- b) Routing for low delay
- c) Multicast routing for one to many data transfer
- d) Transport layer protocol for end to end reliable data transfer
- e) Pricing Schemes
- f) Quality of Service Provisioning
- g) Self-Organization
- h) Security
- i) Addressing and Service Discovery
- j) Energy Management
- k) Scalability
- l) Deployment considerations

4. The difference between cellular and ad hoc network is:-

<i>Criteria</i>	<i>Cellular Network</i>	<i>Ad Hoc Network</i>
Cost Effectiveness	high network maintenance cost	Low network maintenance cost due to self-organisation in network
Bandwidth Usage	easy bandwidth reservation	Access needs complex access control protocol
	Bandwidth can be guaranteed	Shared radio channel hence variable bandwidth

Answers to CYP-2

1. A wireless sensor network is network of communicating devices called nodes over wireless channels and having some sensing capability. These nodes collect the information like temperature, humidity, pressure, noise level etc. and then share it with the base node which then transmits this information to the gateway.
2. A small inexpensive monitoring device with some sensing and computing capability and limited battery to monitor a target environment. The node collects the target information and shares with other nodes wirelessly. In general a node consists of a transducer which converts physical quantities to the voltage levels which are communicated to other nodes/ devices to a relatively short distance.
3. The differences are as following

<i>Single hop</i>	<i>Multi hop</i>
Directly connected with base station	Intermediate devices exists in connection
Uses less energy (only when the geographical distance is small)	Requires more energy
Used in small sensing areas	Used in large sensing regions

4. Different topologies in WSNs are Mesh, Star, Ring, Circular, Bus and Grid topology.

Answers to CYP-3

1. Things in IoT are any physical thing which is of the interest or use to the user. This may include a chair, a toaster, a water bottle etc. These can be converted to smart objects by placing a sensor on them which enables them to communicate and share data with other devices. The characteristics of IoT are Heterogeneity, Dynamic, Interconnected and scaling.

2. Difference of IoT and M2M communication is as following

<i>M2M – Machine 2 Machine</i>	<i>IoT – Internet of Things</i>
Point to point connection establishment	Devices are connected through the network and also supports connecting to global cloud networks.
Limited amount of intelligence	Decision making is enabled
Makes use of internet protocols like- HTTP, FTP, etc.	Makes use of traditional communication protocols
Generally may not rely on internet connection	Generally Rely on internet connection
Less scalable	Highly scalable

3. Short notes on the given topics.

ZigBee

It is a wireless technology based on IEEE 802.15.4 used to address needs of low-power and low-cost IoT devices. It is used to create low cost, low power, low data rate wireless ad-hoc networks. It is resistant to unauthorized reading and communication errors but provides low throughput. It is easy to install, implement and supports a large number of nodes to be connected. It can be used for short range communications only.

NFC

Near Field Communication (NFC) is a protocol used for short distance communication between devices. It is based on RFID technology but has a lower transmission range (of about 10 cm). It is used for identification of documents or objects. It allows contact less transmission of data. It has shorter setup time than Bluetooth and provides better security.

Bluetooth

It is one of the widely used types of wireless PAN used for short range transmission of data. It makes use of short range radio frequency. It provides data rate of approx 2.1 Mbps and operates at 2.45GHz. It is capable of low cost and low power transmission for short distances. Its initial version 1.0 supported upto 732kbps speed. Its latest version is 5.2 which can work upto 400m range with 2 Mbps data rate.
